Smart City: Does Weather Affect Traffic?

Executive summary

Across 32,400 cleaned hourly sensor readings from 15 roads, rain was associated with a 8.1% lower mean speed (Cohen's d=-0.50; 95% CI -0.53 to -0.47). After controlling for time of day, weekday, and road, each additional mm/hour was associated with 0.97 km/h lower speed (95% CI -1.02 to -0.93). Effects varied by corridor; prioritize R04, R01, R02, R03, R05. These are observational/synthetic associations, not causal estimates.

Key findings

• Rain vs dry speed: -8.1% mean change (95% bootstrap CI -8.6% to -7.7%); Cohen's d=-0.50 (medium); Cliff's delta=-0.27 (95% bootstrap CI -0.29 to -0.24; small). Welch p=<0.001; controlled-H0 decision: Reject H0.

• Controlled rainfall slope: -0.97 km/h per mm/hour (robust HC3 95% CI -1.02 to -0.93; p=<0.001); model R-squared=0.784.

• Weather-group speed test: Kruskal-Wallis H=1320.8, p=<0.001, epsilon-squared=0.041 (small); Dunn pairwise results use Bonferroni correction.

• Rain-related volume change: -7.2% (95% bootstrap CI -8.0% to -6.3%); Cohen's d=-0.21; Welch p=<0.001; decision: Reject H0_volume.

• Road effects are heterogeneous: road-by-rain interaction F=261.7, p=<0.001; observed drop spread=20.2 percentage points. The largest peak-period stratified drop was 11.6% in Evening Peak (Weekend); high-sensitivity roads: R04.

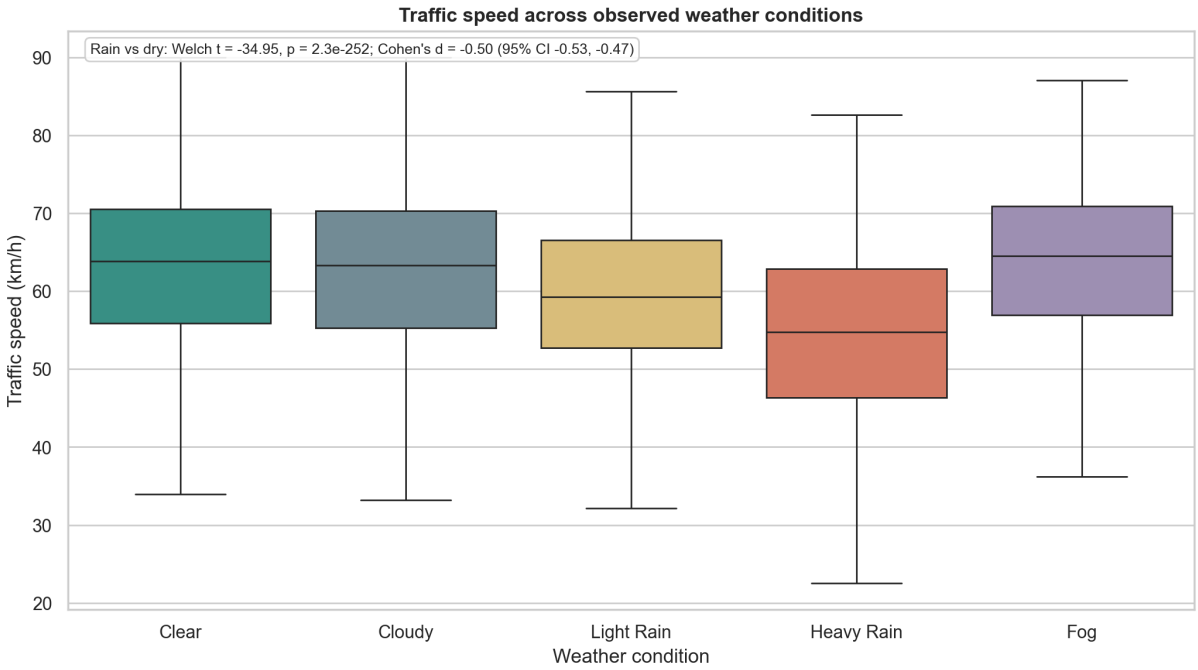
• The largest estimated rain-associated speed drops are R04 (20.3%, d=-1.46, FDR q=2.07e-102), R01 (19.9%, d=-1.50, FDR q=1.05e-104), R02 (18.5%, d=-1.37, FDR q=3.8e-93), R03 (17.4%, d=-1.30, FDR q=8.58e-90), R05 (7.1%, d=-0.55, FDR q=9.28e-24).

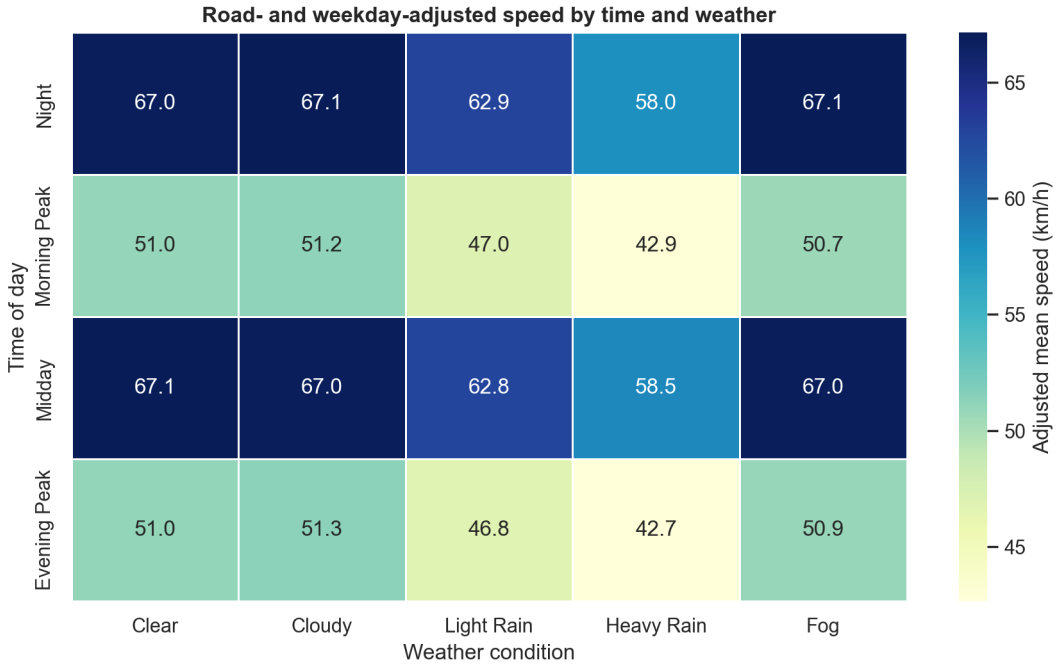
• The held-out decision tree achieved ROC-AUC=0.945, F1=0.149, recall=0.093; without volume, ROC-AUC remained 0.945 but default-threshold recall was 0.000. Logistic regression ROC-AUC=0.953, majority baseline ROC-AUC=0.500. Tree CV ROC-AUC=0.935 +/- 0.009.

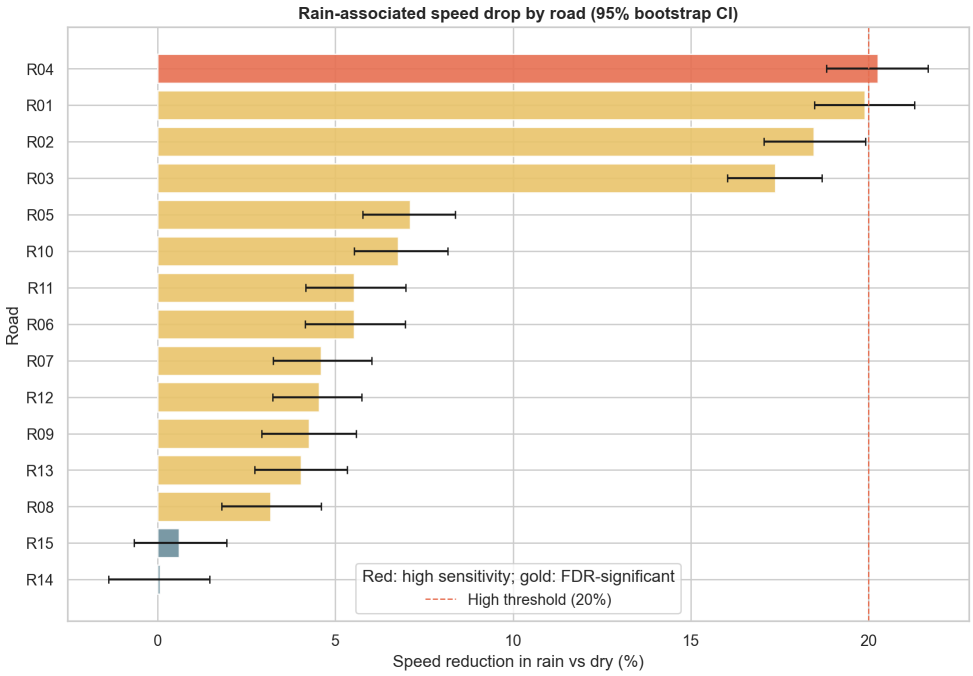
Model evaluation

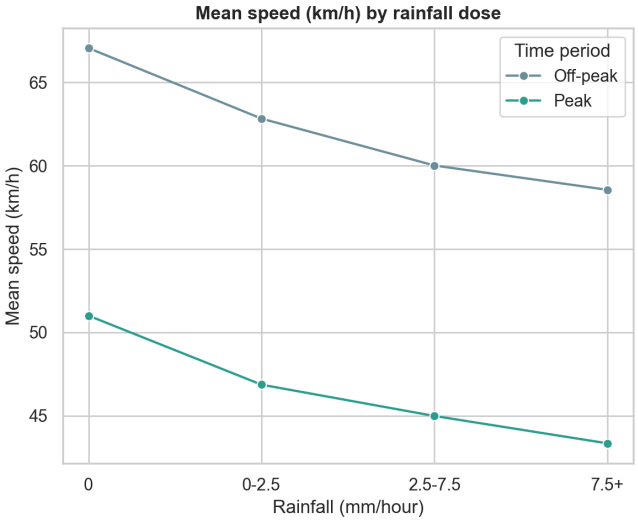
Decision tree: ROC-AUC 0.945, F1 0.149; logistic regression ROC-AUC 0.953.

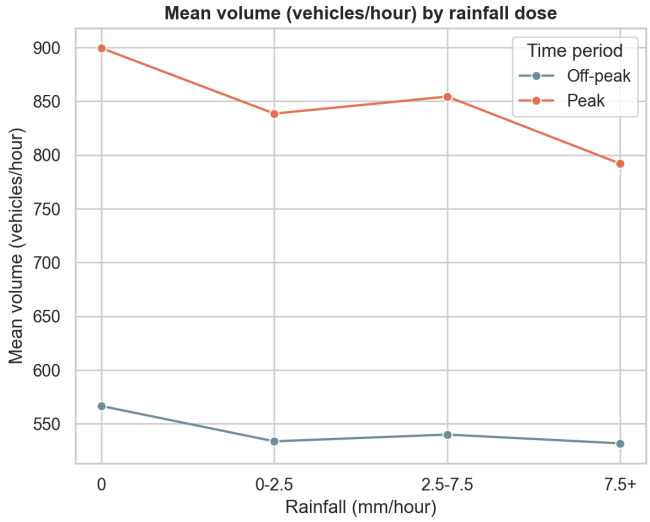
Visual evidence

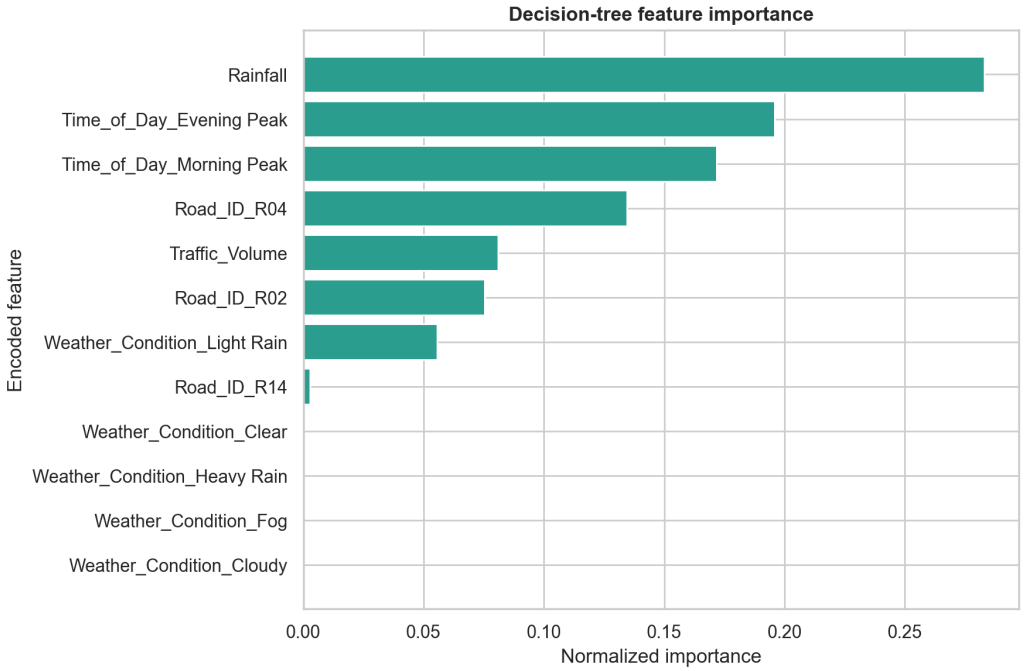


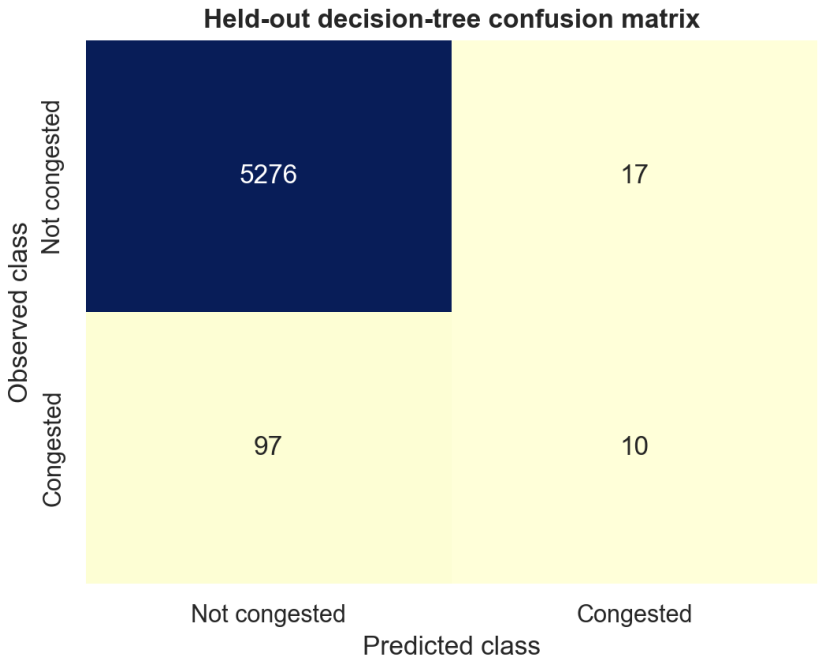



Rain-associated speed drop by road (95% bootstrap CI)
R04
R01
R02
R03
R05
R10
R11
R06
R07
R12
R09
R13
R08
R15
R14
Road
Red: high sensitivity; gold: FDR-significant
High threshold (20%)
0
5
10
15
20
Speed reduction in rain vs dry (%)


Mean speed (km/h) by rainfall dose
Time period
Off-peak
Peak
65
60
55
50
45
Mean speed (km/h)
0
0-2.5
2.5-7.5
7.5+
Rainfall (mm/hour)


Mean volume (vehicles/hour) by rainfall dose
Time period
Off-peak
Peak
900
850
800
750
700
650
600
550
Mean volume (vehicles/hour)
0
0-2.5
2.5-7.5
7.5+
Rainfall (mm/hour)


Decision-tree feature importance
Rainfall
Time_of_Day_Evening Peak
Time_of_Day_Morning Peak
Road_ID_R04
Traffic_Volume
Road_ID_R02
Weather_Condition_Light Rain
Road_ID_R14
Weather_Condition_Clear
Weather_Condition_Heavy Rain
Weather_Condition_Fog
Weather_Condition_Cloudy
Encoded feature
0.00
0.05
0.10
0.15
0.20
0.25
Normalized importance


Held-out decision-tree confusion matrix
5276
17
97
10
Not congested
Congested
Observed class
Not congested
Congested
Predicted class

Practical action plan

Action	Roads / area	Priority	Timeline
Inspect drainage, underpass pumps, and flood paths	R04, R01, R02, R03, R05	High	0-6 months
Pilot rain-triggered adaptive signal timing	High-sensitivity arterials; peak approaches	High	3-9 months
Deploy dynamic speed limits and variable message signs	High- and Moderate-sensitivity corridors	High	3-6 months
Offer rerouting and corridor alerts	R04, R01, R02	Med	3-6 months
Staff incident response for peak-hour rain events	Citywide peak periods; deploy first to ranked roads	Med	Next wet season
Track post-deployment KPIs	Citywide and by road	High	Monthly; evaluate after 6 months

Limitations

The data are synthetic in this run; results demonstrate the reproducible method, not a real city's traffic response. Observational association is not causation. Congestion threshold is a design choice; accidents, roadworks, events, visibility, and route choice are not modeled.